Assignment Copy

Database managment System

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Assignment question:

Chapter 1

1) Define a Database and DBMS
2) Explain the primary purpose of using a database system over a traditional file-based system with a suitable example.
3) Describe the different types of Database users (naive, sophisticated, specialized, etc) and how they interact with the system.
4) write down advantages and disadvantages of the flat file system.
5) Discuss the various areas in which database system is used.
6) who is ~~DBA~~ DBA? List the various responsibilities of DBA.

Solutions

1) Database:

→ Database is an organized collection of releated data that is stored electronically so it can be easily accessed, managed and updated.

example

A school database may store:

- Student names
- Roll numbers
- marks
- Attendence

DBMS (Database managment System)

→ A DBMS is a software that is used to create, manage Store, retrieve, and control database. It acts as an interface between the user and the database.

example

mySQL, oracle, PostgreSQL

2) Explain the primary purpose of using a database System over a traditional file-based system with a suitable example.

→ The main purpose of using a database system is to store and manage data efficiently, secure and Systematically. A database system overcomes many Problems found in traditional file-based systems such as data-redundancy, Inconsistency, and difficulty in ~~as~~ accessing data.

Traditional File Base System

→ In a traditional file system, data is stored in seperate fille, Each department or user may maintains its own files independently.

Problems of file based system:

1) Data Redundancy

→ Same data is stored multiple times in different files.

2) Data inconsistency

→ If data is updated in one file but not in others, Incorrect information may occurs

3) Difficult Data Access

→ Finding and updating data becomes time-consuming

4) Poor Security

→ unauthorized user may access files easily

5) No proper Backup and Recovery

→ Data may be lost permanently if files are damaged.

3) Describe the different type of database users (Naive, sophisticated, specialized, etc) and How they interact with the system.

→ Different types of users interact with a database system according to their needs and techinical knowledge. The main types of database users are described below.

1) Naive users (Parametric users)

→ Naive users are users who have little or no knowledge about database system. They are predefined appication to access the database.

Interaction with the system

→ They interact through simple forms, menus or graphical interfaces without writing database queries.

e.g Bank customers using ATM machine

Students checking results online

Railway ticket booking users.

2) Sophisticated users

→ Sophisticated users have good knowledge of database and can write complex queries using SQL or other database languages.

Interact with the system

→ They directly interact with the database using query languages and analysis tools.

examples:-

Data analysts	Software eg
Engineers	my SQl
Researchers	PostgreSQL.

3) Specialized users.

→ Specialized users are experts who develop special database application for scientific research, artificail intelligence, multimedia system or engineering purposes.

Interaction with System:

→ They use advanced programming languages and special database application to perform complex tasks.

e.g AI developers

scientists

CAD system designers

4) write down advantages and Disadvantages of flat file system.

→ A flat file system is a traditional way of storing data in a single file (or multiple separate files) without any relationship between them. It does not uses a database structure or a DBMS.

Advantage of flat file system

1) simple to understand
→ Easy to create and manage because data is stored in simple files

2) Low cost
→ No need for special software like a database mangment system, so, it is cheaper.

3) Easy for ~~on~~ small Data
→ Suitable for small organizations or personal use where data is limited.

4) fast for small operations
→ for small datasets accessing data can be quick.

Disadvantage of flat file system:

1) Data Redundancy
→ Small data may be stores multiple times in different files.
2) Data Inconsistency
→ If one file is updated and other are not, so data become incorrect.
3) Difficut Data Access
→ Finding specific data becomes hard as data grows.
4) No Data security
→ Anyone who accesses the file can see or modify data
5) No Data Relationship.
→ Files are independent and cannot be linked easily.
6) No Backup and Recovery system
→ Data loss can happen easily if a files is damaged.
7) Pour scalability
→ Not suitable for large organizations or ~~to~~ big data.

5) Discuss the various areas in which database system is used.

→ A database system is used in many fields to store, manage and retrieve large amount of data efficiently. Different organizations use database systems for fast processing, security and ~~Poor~~ proper managment of information.

The major areas where data base system are used :-

1) Banking System

→ Bank use database System to marge costumer accounts, transaction, loans and ATM operations

2) Education Sector

→ School, College and universities use database to maintain students and academic records.

3) Hospital and Healthcare

→ Hospital use database System to store Patients and medical informations.

4) Railway and Airline Reservation System

→ Database System help manage ticket booking and passenger records

5) Library managment System

→ Libraries use database to organize ~~boo~~ books and member information.

6) Who is DBA? List the various responsibilities of DBA.

→ A Database Administrator (DBA) is a person who manges, control and maintains a database System. The DBA is responsible for ensuring that the database operates efficiently, securely and reliably.

The DBA acts as the central authority of the database System.

Responsibilities of DBA :

1) Database installation and configuration
→ The DBA installs and configures the database mangment system and related software.

2) Database security managment
→ The DBA protects the database from unauthorized access by
- creating user accounts
- giving permissions
- managing passwords

3) Backup and Recovery
→ The DBA performs regular backup and restores data if any failure or data loss occurs.

4) Performance monitoring
→ The DBA monitors database performance and improve system speed and efficiency

5) Troubles shooting and Problem solving
→ The DBA identifies and fixes database errors, crashes or technical issues

completed unit 1 assignment

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